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ENERGY STORAGE BOX

Abstract

The present application provides an energy storage box, including a box body, a battery pack, a power conversion system, and a thermal management system. The box body includes a battery cabin, the battery pack and the power conversion system are located in the battery cabin. The thermal management system has a first loop and a second loop. The first loop dissipates heat of a cooling liquid in a first liquid path of the battery pack through a refrigerant. The second loop is in communication with a second liquid path of the power conversion system so as to dissipate heat of the power conversion system by a cooling liquid. The cooling liquid in the second liquid path may also flow to the first liquid path through a solenoid valve so as to form compensation for heat dissipation of the first liquid path.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Chinese Patent Application No. 202410268824.0, filed on Mar. 8, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the technical field of thermal management of energy storage boxes, and specifically relates to an energy storage box.

BACKGROUND

[0003] In the technical fields of micro-grids or new energy power plants such as wind power plants, photovoltaic power plants, and the like, it is generally necessary to dispose an energy storage box so as to satisfy mobile power requirements.

[0004] The energy storage box includes a box body, a battery pack, a power conversion system (PCS), and a temperature control system. The PCS and the battery pack are correspondingly disposed so as to monitor charging and discharging processes of a battery. Since working temperatures of the PCS and the battery pack are different, the temperature control system performs heat exchange on the PCS in an air-cooling manner, and performs heat exchange on the battery pack in a liquid-cooling manner, so that the battery pack and the PCS work stably.

[0005] However, both working manners and control manners of air-cooling and liquid-cooling are different, which reduces a thermal management efficiency of the energy storage box.

SUMMARY

[0006] The present application provides an energy storage box, which can improve the thermal management efficiency of the energy storage box and reduce the energy consumption at the same time.

[0007] In order to achieve the above objects, the present application adopts the following technical solutions: [0008] the present application provides an energy storage box, including a box body, a battery pack, a power conversion system and a thermal management system; where the box body includes a battery cabin, the battery pack and the power conversion system are located in the battery cabin, the power conversion system is electrically connected with the battery pack, the battery pack has a first liquid path, and the power conversion system has a second liquid path; [0009] where the thermal management system is partially located outside the battery cabin, the thermal management system includes a first loop, a second loop and a monitoring unit; the first loop includes a first heat exchanger, a compressor and a second heat exchanger; the first heat exchanger, the compressor and the second heat exchanger are communicated head-to-tail in sequence, the first heat exchanger and the first liquid path are correspondingly disposed; a refrigerant flows in the first loop, the compressor is configured to control the refrigerant to flow in the first loop so as to dissipate heat of a preset liquid in the first liquid path by the refrigerant; [0010] where the second loop includes a radiator and a solenoid valve, the second liquid path is in communication with the radiator, the first liquid path and the radiator are connected in parallel to the second liquid path, and the solenoid valve is disposed between the second liquid path and the first liquid path; [0011] where the monitoring unit includes a temperature sensor and a controller, the temperature sensor is electrically connected with the controller, the temperature sensor is configured to monitor a temperature outside the box body, the controller is electrically connected with the solenoid valve and the compressor; the controller is configured to drive the solenoid valve when the temperature is equal to a preset temperature value, so that a part of the preset liquid in the

second liquid path flows through the first liquid path.

[0012] The energy storage box provided by the present application implements the unification of heat dissipation manners of the battery pack and the power conversion system through the thermal management system, and improves the integration level of the thermal management system of the energy storage box. Meanwhile, working states of the first loop and the second loop, and communication states between the second liquid path and the first liquid path are adjusted by the controller according to the temperature outside the box body, so that the preset liquid in the second liquid path flows into the first liquid path of the battery pack. In this way, the preset liquid in the second liquid path and the preset liquid in the first liquid path together dissipate heat to the battery pack and the power conversion system, and at this time, the rotation speed of the compressor may be gradually reduced, and the energy consumption of heat dissipation of the first loop is gradually reduced. Furthermore, compared with the prior art, the air-cooling heat dissipation structure of the power conversion system is omitted, and the space occupation ratio of the overall structure of the energy storage box is reduced, further improving the energy density of the energy storage station.

[0013] As a possible embodiment, when the temperature is less than the preset temperature value, the controller controls the refrigerant to stop flowing, and drives the solenoid valve, so that a part of the preset liquid in the second liquid path flows through the first liquid path; [0014] the preset temperature value is 0 degrees Celsius.

[0015] As a possible embodiment, there are at least two radiators, and the at least two radiators are connected in parallel to the second liquid path; [0016] there are at least two solenoid valves, one of the at least two solenoid valves and one of the at least two radiators are correspondingly disposed, and one of the at least two solenoid valves is connected in series between the second liquid path and corresponding radiator.

[0017] As a possible embodiment, the first loop further includes an oil separator and a gas-liquid separator, the oil separator is disposed between the compressor and the second heat exchanger; and the compressor, the oil separator and the second heat exchanger are sequentially communicated; [0018] the gas-liquid separator is disposed between the first heat exchanger and the compressor; and the first heat exchanger, the gas-liquid separator and the compressor are sequentially communicated.

[0019] As a possible embodiment, there are at least two second heat exchangers, and the at least two second heat exchangers are connected in parallel between the compressor and the first heat exchanger.

[0020] As a possible embodiment, the box body further includes a device control cabin, the device control cabin and the battery cabin are disposed at intervals along a length direction of the box body, the compressor is located in the device control cabin, and a top of the device control cabin is in communication with the outside of the box body.

[0021] As a possible embodiment, the second heat exchanger and the radiator are disposed at intervals along the length direction of the box body at an outer top of the box body.

[0022] As a possible embodiment, the thermal management system further includes a first draught fan and a second draught fan, the first draught fan and the second heat exchanger are correspondingly disposed, the first draught fan is connected with a top of the box body, and the first draught fan is configured to dissipate heat of the second heat exchanger; [0023] the second draught fan and the radiator are correspondingly disposed, the second draught fan is connected with the top of the box body, and the second draught fan is configured to dissipate heat of the radiator.

[0024] As a possible embodiment, at least one of the first draught fan and the second draught fan includes a housing and a fan, the housing has an accommodating cavity, a top of the accommodating cavity has a first vent, and the fan is located in the accommodating cavity so as to discharge air in the accommodating cavity along the first vent; [0025] a side wall of the housing is provided with a second vent, the second vent is in communication with the accommodating cavity; a top of the housing is provided with a third vent, the third vent and the second vent are

correspondingly disposed, and the third vent is in communication with the second vent.
[0026] As a possible embodiment, the first heat exchanger is a plate type heat exchanger.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0027] In order to illustrate the technical solutions in the embodiments of the present application or in the prior art more clearly, the following briefly introduces the accompanying drawings required for describing the embodiments or the prior art. Apparently, the accompanying drawings in the following description are some embodiments of the present application, and for those skilled in the art, other drawings may also be obtained according to these drawings without creative efforts.

[0028] FIG. 1 is a schematic structural diagram of an energy storage box provided by an embodiment of the present application.

[0029] FIG. 2 is a schematic diagram of an internal structure of an energy storage box provided by an embodiment of the present application.

[0030] FIG. 3 is a working principle diagram of a thermal management system of an energy storage box provided by an embodiment of the present application.

[0031] FIG. 4 is a top view of an energy storage box provided by an embodiment of the present application.

[0032] FIG. 5 is a front view of an energy storage box provided by an embodiment of the present application.

DESCRIPTION OF THE REFERENCE NUMBERS

[0033] **100**—energy storage box; [0034] **1**—box body; **11**—device control cabin; [0035] **2**—battery pack; **21**—first liquid path; [0036] **3**—power conversion system; **31**—second liquid path; [0037] **4**—thermal management system; **41**—first heat exchanger; **42**—compressor; [0038] **43**, **43a**, **43b**—second heat exchanger; **44**, **44a**, **44b**—radiator; [0039] **45**, **45a**, **45b**—solenoid valve; **46**—oil separator; **47**—gas-liquid separator; [0040] **48**—first draught fan; **49**—second draught fan; [0041] **491**—first vent; **492**—second vent; **493**—third vent.

DESCRIPTION OF EMBODIMENTS

[0042] In order to make the objectives, technical solutions, and advantages of the embodiments of the present disclosure clearer, the following clearly and completely describes the technical solutions in the embodiments of the present disclosure with reference to the accompanying drawings in the embodiments of the present disclosure. Apparently, the described embodiments are merely a part rather than all of the embodiments of the present disclosure. Based on the embodiments of the present disclosure, all other embodiments obtained by persons of ordinary skill in the art without creative efforts shall belong to the scope of protection of the present disclosure. Without conflict, the following embodiments and features in the embodiments may be combined with each other.

[0043] In the technical fields of micro-grids or new energy power plants such as wind power plants, photovoltaic power plants, and the like, it is generally necessary to dispose an energy storage box so as to satisfy mobile power requirements.

[0044] The energy storage box includes a box body, a battery pack, a power conversion system (PCS), and a temperature control system. The PCS and the battery pack are correspondingly disposed so as to monitor charging and discharging processes of a battery. Since working temperatures of the PCS and the battery pack are different, the temperature control system performs heat exchange on the PCS in an air-cooling manner, and performs heat exchange on the battery pack in a liquid-cooling manner, so that the battery pack and the PCS work stably.

[0045] However, both working manners and control manners of air-cooling and liquid-cooling are different, which reduces a thermal management efficiency of the energy storage box.

[0046] In order to overcome the defects in the prior art, the present application provides an energy storage box, including a box body, a battery pack, a power conversion system, and a thermal management system. Where, the box body includes a battery cabin, the battery pack and the power conversion system are located in the battery cabin. The thermal management system has a first loop and a second loop. The first loop dissipates heat of a cooling liquid in a liquid-cooling loop of the battery pack through flowing refrigerant. A cooling liquid in the second loop flows in a liquid path of the power conversion system so as to dissipate heat of the power conversion system. When a temperature of the environment outside the box body is at a specific preset temperature value, the cooling liquid in the liquid path of the power conversion system may also flow to a liquid path of the battery pack through a solenoid valve, so as to reduce the working rotation speed of a compressor in the first loop, forming heat dissipation compensation for the liquid path of the battery pack. In this way, the thermal management system achieves the unification of heat dissipation manners in the energy storage box, and improves the integration level of the energy storage box, and reduces the overall structure of the energy storage box, and reduces the space occupation ratio of the energy storage box, thereby improving the energy density of a station. Meanwhile, the liquid path of the power conversion system compensates for heat dissipation of the liquid path of the battery pack, which can reduce the energy consumption of the first loop, thereby improving the energy density and energy storage efficiency of the station.

[0047] The following describes the content of the present disclosure in detail with reference to the accompanying drawings, so that those skilled in the art can understand the content of the present disclosure in more detail.

[0048] FIG. 1 is a schematic structural diagram of an energy storage box provided by an embodiment of the present application. FIG. 2 is a schematic diagram of an internal structure of an energy storage box provided by an embodiment of the present application. FIG. 3 is a working principle diagram of a thermal management system of an energy storage box provided by an embodiment of the present application. FIG. 4 is a top view of an energy storage box provided by an embodiment of the present application. FIG. 5 is a front view of an energy storage box provided by an embodiment of the present application. As shown in FIG. 1 to FIG. 5, an embodiment of the present application provides an energy storage box **100**, including a box body **1**, a battery pack **2**, a power conversion system **3** and a thermal management system **4**, where the box body **1** includes a battery cabin, the battery pack **2** and the power conversion system **3** are located in the battery cabin, and the power conversion system **3** is electrically connected with the battery pack **2**. The battery pack **2** has a first liquid path **21**, and the power conversion system **3** has a second liquid path **31**.

[0049] The thermal management system **4** is partially located outside the battery cabin, and the thermal management system **4** includes a first loop, a second loop and a monitoring unit. The first loop includes a first heat exchanger **41**, a compressor **42** and a second heat exchanger **43**. The first heat exchanger **41**, the compressor **42** and the second heat exchanger **43** are communicated head-to-tail in sequence. The first heat exchanger **41** and the first liquid path **21** are correspondingly disposed. A refrigerant flows in the first loop, the compressor **42** is configured to control the refrigerant to flow in the first loop so as to dissipate heat of a preset liquid in the first liquid path **21** by the refrigerant.

[0050] The second loop includes a radiator **44** and a solenoid valve **45**, the second liquid path **31** is in communication with the radiator **44**, the first liquid path **21** and the radiator **44** are connected in parallel to the second liquid path **31**, and the solenoid valve **45** is disposed between the second liquid path **31** and the first liquid path **21**.

[0051] The monitoring unit includes a temperature sensor and a controller. The temperature sensor is electrically connected with the controller. The temperature sensor is configured to monitor a temperature outside the box body **1**. The controller is electrically connected with the solenoid valve **45** and the compressor **42**, and the controller is configured to drive the solenoid valve **45** when the

temperature is equal to a preset temperature value, so that a part of the preset liquid in the second liquid path **31** flows through the first liquid path **21**.

[0052] In the present embodiment, the thermal management system **4** implements the unification of heat dissipation manners of the battery pack **2** and the power conversion system **3**, and improves the integration level of the thermal management system **4** of the energy storage box **100**.

Meanwhile, working states of the first loop and the second loop, and communication states between the second liquid path **31** and the first liquid path **21** are adjusted by the controller according to the temperature outside the box body **1**, so that the preset liquid in the second liquid path **31** flows into the first liquid path **21** of the battery pack **2**. In this way, the preset liquid in the second liquid path **31** and the preset liquid in the first liquid path **21** together dissipate heat to the battery pack **2** and the power conversion system **3**, and at this time, the rotation speed of the compressor **42** may be gradually reduced, and the energy consumption of heat dissipation of the first loop is gradually reduced. Furthermore, compared with the prior art, the air-cooling heat dissipation structure of the power conversion system **3** is omitted, and the space occupation ratio of the overall structure of the energy storage box **100** is reduced, further improving the energy density of the energy storage station.

[0053] For example, the battery pack **2** and the power conversion system **3** in the present embodiment are located in the battery cabin, and the power conversion system **3** and the battery pack **2** are arranged at intervals along a height direction of the box body **1**. It should be understood that the power conversion system **3** is located at a bottom of the box body **1**. The battery pack **2** is provided with the first liquid path **21**, and the power conversion system **3** is provided with the second liquid path **31**. Both the first liquid path **21** and the second liquid path **31** are in communication with the thermal management system **4**. The heat dissipation and cooling of the preset liquids in the first liquid path **21** and the second liquid path **31** are performed through the thermal management system **4**. In this way, heat dissipation manners of the battery pack **2** and the power conversion system **3** are both implemented by the thermal management system **4**, which changes the air-cooling heat dissipation manner of the power conversion system **3** and omits the air-cooling heat dissipation structure, so that both the battery pack **2** and the power conversion system **3** can be integrated in the battery cabin, thereby reducing the overall structure of the energy storage box **100** and reducing the space occupation ratio of the energy storage box **100**. It should be understood that, in a station, the space occupation ratio of an individual energy storage box **100** decreases, which means that the total number of energy storage boxes **100** in the station increases, so that the energy density of the station and the total energy storage quantity of the station can be improved.

[0054] Further, the thermal management system **4** includes the first loop. The first loop performs heat exchange on the preset liquid in the first liquid path **21** of the battery pack **2** by the refrigerant, so as to implement heat dissipation of the battery pack **2**, thereby simultaneously ensuring a working temperature of the battery pack **2** and ensuring security of the energy storage box **100**. Specifically, the first loop includes the first heat exchanger **41**, the compressor **42**, and the second heat exchanger **43**. The first heat exchanger **41**, the compressor **42**, and the second heat exchanger **43** are communicated in sequence to form a flow path of the refrigerant. Where, the first heat exchanger **41** and the first liquid path **21** are correspondingly disposed. A blade of the compressor **42** rotates, so as to compress the refrigerant into a high-temperature and high-pressure gaseous refrigerant, and discharge the high-temperature and high-pressure gaseous refrigerant to the second heat exchanger **43**. Under the action of the second heat exchanger **43**, the high-temperature and high-pressure gaseous refrigerant is converted into a low-temperature and high-pressure liquid to flow to the first heat exchanger **41** in the first loop. When the preset liquid in the first liquid path **21** of the battery pack **2** flows through the first heat exchanger **41**, the preset liquid in the first liquid path **21** exchanges heat with the low-temperature liquid refrigerant in the first heat exchanger **41**. The refrigerant absorbs heat of the preset liquid and is converted into gas, and then flows to the

compressor **42** to complete a next refrigeration cycle. In this process, the preset liquid in the first liquid path **21** is implemented the decrease of the temperature under the action of the refrigerant, and continues to circulate in the first liquid path **21**, and forms heat conduction with the battery pack **2** to take away the heat of the battery pack **2**, achieving heat dissipation and cooling of the battery pack **2**, and completing heat conduction with the refrigerant again in the next cycle. Repeating in this way, by means of the cooperation between the first loop and the first liquid path **21**, the thermal management of the battery pack **2** is completed, so that the working temperature of the battery pack **2** is maintained stable.

[0055] On this basis, the thermal management system **4** further includes the second loop, and the preset liquid flows in the second loop. The second loop is in communication with the second liquid path **31** of the power conversion system **3** so as to form a path of the preset liquid. The preset liquid forms heat conduction with an external environment of the box body **1** in a flowing process, and circulates in the second loop and the second liquid path **31** to dissipate heat to the power conversion system **3**. Specifically, the second loop includes the radiator **44**, and the radiator **44** is in communication with the second liquid path **31**. In the flowing process, the preset liquid in the second liquid path **31** forms heat conduction with the power conversion system **3** to dissipate heat of the power conversion system **3**. The preset liquid continues to flow to the radiator **44**, and the temperature of the preset liquid at the radiator **44** is higher than the temperature of external environment, so that heat can be released to the external environment through the radiator **44**, and the temperature of the preset liquid decreases, and the preset liquid continues to flow into the second liquid path **31** and dissipates heat of the power conversion system **3** again. Repeating in this way, the preset liquid in the second liquid path **31** can continuously dissipate heat of the power conversion system **3**, so as to maintain the working temperature of the power conversion system **3**.

[0056] Further, the second loop also includes the solenoid valve **45**. The solenoid valve **45** is disposed between the second liquid path **31** and the first liquid path **21**, and the solenoid valve **45** can control the communication state between the second liquid path **31** and the first liquid path **21**. When the solenoid valve **45** is closed, the first loop dissipates heat from the preset liquid in the first liquid path **21** of the battery pack **2**, and the preset liquid in the second liquid path **31** of the power conversion system **3** is dissipated heat through the radiator **44** in the second loop, that is to say, heat dissipation control between the battery pack **2** and the power conversion system **3** is independent from each other. When the solenoid valve **45** is opened, the second liquid path **31** is in communication with the first liquid path **21** at this time, and when the preset liquid in the second liquid path **31** of the power conversion system **3** flows, a part of the preset liquid flows into the second loop, another part of the preset liquid flows into the first liquid path **21** and circulates in the first liquid path **21** to dissipate heat of the battery pack **2**, thereby forming heat dissipation compensation of the preset liquid in the second liquid path **31** to the preset liquid in the first liquid path **21**. Meanwhile, the opening degree of the solenoid valve **45** is controllable, so that the flow rate of the preset liquid flowing from the second liquid path **31** to the first liquid path **21** can be controlled. In this way, by controlling the flow rate of the preset liquid flowing from the second liquid path **31** to the first liquid path **21**, the degree of heat dissipation compensation of the second liquid path **31** to the first liquid path **21** can be controlled. On the basis of satisfying the heat dissipation of the battery pack **2** and the power conversion system **3**, the energy consumption of the thermal management system **4** is controlled so as to reduce the energy consumption of the thermal management system **4**.

[0057] In the present embodiment, an opening-and-closing control of the solenoid valve **45** is achieved by the monitoring unit. Specifically, the monitoring unit includes the temperature sensor and the controller. The temperature sensor is disposed on the box body **1** and monitors the temperature of the environment outside the box body **1**. The temperature sensor is electrically connected with the controller, and the controller is electrically connected with the solenoid valve **45**. When the temperature outside the box body **1**, monitored by the temperature sensor, is equal to

the preset temperature value, the controller may control the solenoid valve **45** to be opened, so that the second liquid path **31** is in communication with the first liquid path **21**. In this way, a part of the preset liquid in the second liquid path **31** may flow from the second liquid path **31** to the first liquid path **21**, and mix with an original preset liquid in the first liquid path **21**, and then circulate in the first liquid path **21** to dissipate heat of the battery pack **2**. In this way, due to the introduction of the preset liquid in the second liquid path **31**, the temperature of the preset liquid in the first liquid path **21** decreases, and it should be understood that, the relative working pressure of heat dissipation of the first loop decreases, and the compressor **42** may gradually slow down the operation, thereby reducing the energy consumption of the first loop, so as to reduce the overall energy consumption of the thermal management system **4**.

[0058] It should be noted that, the preset liquid in the present embodiment is a cooling liquid, and a substance with a higher specific heat capacity of water, may be selected, so that a heat conduction effect of the preset liquid, circulating in the first liquid path **21**, the second liquid path **31** and the second loop, is more significant. In addition, the solenoid valve **45** in the present embodiment is a three-way solenoid valve **45**, and the preset temperature value is 0 to 20 degrees Celsius.

[0059] As a possible embodiment, when the temperature is less than the preset temperature value, the controller controls the refrigerant to stop flowing, and drives the solenoid valve **45**, so that a part of the preset liquid in the second liquid path **31** flows through the first liquid path **21**. The preset temperature value is 0 degrees Celsius. In this way, in the thermal management system **4**, only through the preset liquid in the second liquid path **31** and the first liquid path **21**, the battery pack **2** and the power conversion system **3** are simultaneously dissipated heat. At this time, the first loop may be completely non-working, and only through the second loop, the heat conduction between the preset liquid and the external environment is completed, thereby implementing a heat dissipation requirement of the energy storage box **100**.

[0060] It should be understood that, the battery pack **2** and the power conversion system **3** in the energy storage box **100** have different temperature requirements for maintaining an efficient and stable operation. The temperature required by the battery pack **2** is lower than that of the power conversion system **3**. In the present embodiment, when the temperature of the environment outside the box body **1** is low, the temperature of the preset liquid in the first liquid path **21** of the battery pack **2** is higher than the temperature of the environment outside the box body **1**, and a temperature difference is formed, and during the process of flowing of the preset liquid, the heat conduction is naturally formed, that is, the preset liquid dissipates heat to the battery pack **2** through cold air outside the box body **1**, and accordingly, the first loop may implement the heat dissipation of the battery pack **2** without working, so as to maintain the normal operation of the battery pack **2**.

[0061] When the temperature of the environment becomes lower and lower, the temperature of the preset liquid in the first liquid path **21** also decreases accordingly, which easily leads to unstable working state of the battery pack **2**. However, the temperature of the preset liquid in the second liquid path **31** of the power conversion system **3** is higher than the temperature of the preset liquid in the first liquid path **21**, at this time, the preset liquid in the second liquid path **31** may be introduced into the first liquid path **21** to preserve heat of the battery pack **2**, so that the thermal management may be performed on the battery pack **2** and the power conversion system **3** only through the second loop. Therefore, in a case that when the temperature of the environment outside the box body **1** is lower than 0 degrees Celsius, in the present embodiment, the controller controls the compressor **42** in the first loop to stop working, so that the refrigerant in the first loop stops flowing and does not exchange heat with the preset liquid in the first liquid path **21**. Meanwhile, the controller opens the solenoid valve **45**, so that the second liquid path **31** is in communication with the first liquid path **21**, a part of the preset liquid in the second liquid path **31** flows into the first liquid path **21**; and another part of the preset liquid circulates in the second loop, after forming heat exchange with the cold air outside the box body **1**, then continues to flow into the second liquid path **31**. In this way, in a low-temperature environment, a thermal management of the energy

storage box **100** may be implemented only through the second loop, and on the basis of ensuring normal operation of the energy storage box **100**, the operating energy consumption of the thermal management system **4** is reduced.

[0062] In order to ensure the heat dissipation efficiency of the second loop under various temperature conditions, in the present embodiment, there are at least two radiators **44**, and at least two radiators **44** are connected in parallel to the second liquid path **31**. There are at least two solenoid valves **45**. One of the at least two solenoid valves **45** and one of the at least two radiators **44** are correspondingly disposed, and one of the at least two solenoid valves **45** is connected in series between the second liquid path **31** and corresponding radiator **44**.

[0063] In the present embodiment, there are two radiators **44**, and the two radiators **44** are connected in parallel to the second liquid path **31**. In this way, when the preset liquid in the second liquid path **31** flows to the second loop, it is divided into two parts, and after passing through the two radiators **44**, returns to the second liquid path **31**. It should be understood that, the preset liquid in the second loop forms heat conduction with the air outside the box body **1** through the two radiators **44**, so as to implement the change of the temperature of the preset liquid, thereby improving the heat dissipation efficiency of the power conversion system **3**.

[0064] Further, in the present embodiment, the solenoid valve **45** and the radiator **44** are correspondingly disposed, that is to say, the solenoid valve **45a** is connected in series to the radiator **44a**, and the solenoid valve **45b** is connected in series to the radiator **44b**; correspondingly, the solenoid valve **45a** is connected in series between the second liquid path **31** and the radiator **44a**, and the solenoid valve **45b** is connected in series between the second liquid path **31** and the radiator **44b**. In other words, two solenoid valves **45** are connected in parallel to the second liquid path **31**. By controlling the relative opening-and-closing states of the solenoid valve **45a** and the solenoid valve **45b**, the heat dissipation efficiency of the second loop may be controlled. Specifically, the solenoid valve **45a** and the solenoid valve **45b** may be alternatively opened, so that only one of the two radiators **44** in the second loop is working, and the heat dissipation requirement of the power conversion system **3** may be satisfied. When the temperature of the preset liquid in the second liquid path **31** increases, and the working of one radiator **44** may no longer satisfy the heat dissipation requirement, two solenoid valves **45** may be simultaneously opened, and correspondingly, the preset liquid in the second liquid path **31** forms a shunt in the second loop, flowing to the radiator **44a** and the radiator **44b** respectively, and the radiator **44a** and the radiator **44b** simultaneously work so as to improve the heat exchange efficiency between the preset liquid in the second loop and the air outside the box body **1**; then the working temperature of the power conversion system **3** is maintained within an appropriate range, thereby ensuring stable operation of the energy storage box **100**.

[0065] Further, in the present embodiment, the solenoid valve **45a** and the solenoid valve **45b** are both connected in series to the first loop, and a parallel relationship is formed between the solenoid valve **45a** and the solenoid valve **45b**. In combination with the above-mentioned embodiments, the solenoid valve **45a** and the solenoid valve **45b** may be alternatively opened, so that the second liquid path **31** is in communication with the first liquid path **21**. For example, when the solenoid valve **45a** is opened, the second liquid path **31** is in communication with the radiator **44a** in the second loop through the solenoid valve **45a**, and the second liquid path **31** is in communication with the first liquid path **21** through the solenoid valve **45a**. In this way, a part of the preset liquid in the second liquid path **31** flows to the first liquid path **21** through the solenoid valve **45a**, then returns to the second liquid path **31**, and at the same time, the preset liquid in the second liquid path **31** is dissipated heat through the radiator **44a** in the second loop. When the solenoid valve **45b** is opened, the working principle of the thermal management system **4** is the same as the principle of only opening the solenoid valve **45a**, which is not repeated herein.

[0066] It should not be difficult to be understood that, when the solenoid valve **45a** and the solenoid valve **45b** are simultaneously opened, the preset liquid in the second liquid path **31** may

flow to the first liquid path **21** through both solenoid valves **45**, and after circulating in the first liquid path **21**, return to the second liquid path **31**; and finally under the action of the radiator **44** of the second loop, the heat exchange between the preset liquid and the air outside the box body **1** is completed, so that the thermal management efficiency of the thermal management system **4** can be controlled.

[0067] When the temperature outside the box body **1** is 0 to 20 degrees Celsius, due to the opening of the solenoid valve **45**, the heat dissipation of the preset liquid in the first liquid path **21** is completed under a combined action of the first loop, the second liquid path **31** and the second loop. It should be understood that, the second liquid path **31** and the second loop form a compensation effect for the heat dissipation of the first liquid path **21**, that is to say, a requirement for the preset liquid in the first liquid path **21** to dissipate heat through the first loop would be reduced; accordingly, the compressor **42** may be appropriately adjusted so as to reduce the rotation speed, so that the energy consumption of the compressor **42** may be correspondingly reduced, so as to achieve the effect of reducing the overall energy consumption of the thermal management system **4**.

[0068] It should be noted that, in the present embodiment, the number of the radiator **44** and the solenoid valve **45** are not limited to the two mentioned in the foregoing embodiments, and the number of the radiator **44** and the solenoid valve **45** may be selected according to a thermal management requirement of the energy storage box **100**, which is not specifically limited in the present embodiment.

[0069] As a possible embodiment, the first loop further includes an oil separator **46** and a gas-liquid separator **47**, the oil separator **46** is disposed between the compressor **42** and the second heat exchanger **43**; and the compressor **42**, the oil separator **46** and the second heat exchanger **43** are sequentially communicated. The gas-liquid separator **47** is disposed between the first heat exchanger **41** and the compressor **42**; and the first heat exchanger **41**, the gas-liquid separator **47** and the compressor **42** are sequentially communicated. In the present embodiment, through the arrangement of the oil separator **46** and the gas-liquid separator **47**, the heat dissipation efficiency of the first loop to the preset liquid in the first liquid path **21** can be improved, meanwhile, the energy consumption of the first loop can be reduced.

[0070] The blades in the compressor **42** rotate to compress the refrigerant and discharge it. It is inevitable that a certain amount of lubricating oil is brought out from the compressor **42**. The lubricating oil has a certain lubrication effect on the flow of the refrigerant, and may also ensure the operation reliability of the first loop at the same time. However, an accumulation of the lubricating oil in the first loop would cause the refrigerant flowing unsmoothly, and the heat dissipation effect of the first loop on the preset liquid in the first liquid path **21** becomes poor; and in order to ensure the heat dissipation effect of the preset liquid in the first liquid path **21**, the power of the compressor **42** would be increased, which leads to an increase of the energy consumption of the first loop. Therefore, in the present embodiment, the oil separator **46** is disposed between the compressor **42** and the second heat exchanger **43**; and the compressor **42**, the oil separator **46** and the second heat exchanger **43** are sequentially communicated. The compressor **42** works to discharge the high-temperature and high-pressure gaseous refrigerant to the oil separator **46**, and the lubricating oil discharged together with the refrigerant also enters the oil separator **46**, they would be separated in the oil separator **46**, and finally only the high-temperature and high-pressure gaseous refrigerant is discharged. The high-temperature and high-pressure gaseous refrigerant forms a rapid heat exchange with the outside of the box body **1** at the second heat exchanger **43**, and at this time, the high-temperature and high-pressure gaseous refrigerant is converted into the low-temperature and high-pressure liquid refrigerant.

[0071] The liquid refrigerant flows to the first heat exchanger **41**, and at the first heat exchanger **41**, completes heat exchange with the preset liquid in the first liquid path **21** of the battery pack **2**. At this time, the liquid refrigerant is converted into a gaseous refrigerant again, and the gaseous

refrigerant completes a next heat dissipation cycle under the action of the compressor **42**. During this process, transitions of the refrigerant of two different forms of gas phase and liquid phase are formed in the first loop. Since components in the first loop communicate with each other, the refrigerant flows in the first loop under the action of the compressor **42**. In order to prevent the liquid refrigerant from entering the compressor **42** and adhering to the blades of the compressor **42**, which causes the effect of the compressor **42** on the gaseous refrigerant to be weakened, in the present embodiment, the gas-liquid separator **47** is connected in series between the first heat exchanger **41** and the compressor **42**; and the first heat exchanger **41**, the gas-liquid separator **47** and the compressor **42** are sequentially communicated. The gas-liquid separator **47** performs gas-liquid separation on the refrigerant flowing from the first heat exchanger **41** to the compressor **42**, so as to ensure that the refrigerant flowing to the compressor **42** is in a gaseous state as far as possible. In this way, the compressing effect of the compressor **42** on the refrigerant may be improved, and then the heat dissipation effect and the working efficiency of the first loop may be improved.

[0072] In order to improve the heat dissipation efficiency of the first loop to the first liquid path **21**, in the present embodiment, there are at least two second heat exchangers **43**, and the at least two second heat exchangers **43** are connected in parallel between the compressor **42** and the first heat exchanger **41**.

[0073] For example, in the present embodiment, there are two second heat exchangers **43**, the second heat exchanger **43a** is in communication with the compressor **42** and the first heat exchanger **41**, the second heat exchanger **43b** is in communication with the compressor **42** and the first heat exchanger **41**, and the second heat exchanger **43a** and the second heat exchanger **43b** are connected in parallel in the first loop. In this way, the high-temperature and high-pressure gaseous refrigerant discharged by the compressor **42** is divided into two paths in the first loop, and the two paths enter the second heat exchanger **43a** and the second heat exchanger **43b**, and perform a rapid heat exchange with the air outside the box body **1** at the second heat exchanger **43a** and the second heat exchanger **43b** respectively, which converts the high-temperature and high-pressure gaseous refrigerant into the low-temperature and high-pressure liquid refrigerant; the liquid refrigerant flows to the first heat exchanger **41**, and performs heat exchange with the preset liquid in the first liquid path **21** at the first heat exchanger **41**, thereby implementing heat dissipation of the preset liquid in the first liquid path **21**. In this way, by arranging the two second heat exchangers **43**, a conversion rate of the refrigerant may be improved, and the working efficiency of the thermal management system is improved.

[0074] It should be noted that, in the present embodiment, the number of the second heat exchanger **43** is not limited to two as mentioned in the foregoing embodiments, and the number of the second heat exchanger **43** may be selected according to a thermal management requirement of the energy storage box **100**, which is not specifically limited in the present embodiment.

[0075] In some embodiments, the box body **1** further includes a device control cabin **11**, the device control cabin **11** and the battery cabin are disposed at intervals along a length direction of the box body **1**, the compressor **42** is located in the device control cabin **11**, and a top of the device control cabin **11** is in communication with the outside of the box body **1**. In this way, a part of components of the thermal management system **4** are arranged outside the battery cabin, so that the temperature of the environment in the battery cabin is suitable, and the working stability of the battery pack **2** and the power conversion system **3** is improved.

[0076] On the basis of the foregoing embodiment, the second heat exchanger **43** and the radiator **44** are disposed at intervals along the length direction of the box body **1** at an outer top of the box body **1**. In this way, the second heat exchanger **43** and the radiator **44** are disposed at the top of the box body **1**, which relatively increases a contact area between the second heat exchanger **43** and the air outside the box body **1**, and between the radiator **44** and the air outside the box body **1**, so that heat exchange rates between the second heat exchanger **43** and the air outside the box body **1**, and

between the radiator **44** and the air outside the box body **1** are increased, thereby improving the heat dissipation efficiency and heat dissipation effect of the thermal management system **4** on the battery pack **2** and the power conversion system **3**.

[0077] As shown in FIG. **1**, FIG. **4** and FIG. **5**, in the present embodiment, two radiators **44** are disposed at intervals along the length direction of the box body **1** on a side away from the device control cabin **11**. Two second heat exchangers **43** are disposed at intervals along the length direction of the box body **1** on a side close to the device control cabin **11**, and the two second heat exchangers **43** are in communication with the compressor **42** in the device control cabin **11**. In this way, the relative distance between the second heat exchanger **43** and the compressor **42** is shortened, so that a flow path of the refrigerant in the first loop is shortened, a thermal loss of the refrigerant in the flowing process is reduced, and the thermal management efficiency of the thermal management system **4** for the battery pack **2** and the power conversion system **3** is further ensured.

[0078] Further, the thermal management system **4** further includes a first draught fan **48** and a second draught fan **49**, the first draught fan **48** and the second heat exchanger **43** are correspondingly disposed, the first draught fan **48** is connected with a top of the box body **1**, and the first draught fan **48** is configured to dissipate heat of the second heat exchanger **43**. The second draught fan **49** and the radiator **44** are correspondingly disposed, the second draught fan **49** is connected with the top of the box body **1**, and the second draught fan **49** is configured to dissipate heat of the radiator **44**. In this way, by arranging the first draught fan **48** and the second draught fan **49**, the heat exchange rates between the second heat exchanger **43** and the air outside the box body **1**, and between the radiator **44** and the air outside the box body **1** are increased.

[0079] As shown in FIG. **1**, FIG. **4** and FIG. **5**, in the present embodiment, the first draught fan **48** and the second heat exchanger **43** are correspondingly disposed, and the first draught fan **48** is disposed at the outer top of the box body **1**. The refrigerant flows in the second heat exchanger **43**, and implements heat exchange with ambient air through the second heat exchanger **43**, and it should be understood that, the temperature of the ambient air of the second heat exchanger **43** increases, and a change of a flow velocity of hot air affects the heat exchange efficiency of the refrigerant at the second heat exchanger **43**. Therefore, in the present embodiment, the first draught fan **48** is started; the first draught fan **48** may blow air to the second heat exchanger **43**, and may also suck out the hot air at the second heat exchanger **43**, in this way, accumulation of the hot air at the second heat exchanger **43** may be reduced, thereby improving the heat dissipation effect and efficiency of the first loop for the preset liquid in the first liquid path **21**.

[0080] Similarly, the second draught fan **49** and the radiator **44** are correspondingly disposed, and the second draught fan **49** is disposed at the outer top of the box body **1**. The preset liquid in the second liquid path **31** flows in the radiator **44** of the second loop, and implements heat exchange with ambient air through the radiator **44**, and it should be understood that, the temperature of the ambient air of the radiator **44** increases, and a change of a flow velocity of hot air affects the heat exchange efficiency of the preset liquid at the radiator **44**. Therefore, in the present embodiment, the second draught fan **49** is started; the second draught fan **49** may blow air to the radiator **44**, and may also suck out the hot air at the radiator **44**, in this way, accumulation of the hot air at the radiator **44** may be reduced, thereby improving the heat dissipation effect and efficiency of the second loop for the preset liquid in the second liquid path **31**.

[0081] Specifically, as shown in FIG. **1**, FIG. **4** and FIG. **5**, at least one of the first draught fan **48** and the second draught fan **49** includes a housing and a fan, the housing has an accommodating cavity, a top of the accommodating cavity has a first vent **491**, and the fan is located in the accommodating cavity so as to discharge air in the accommodating cavity along the first vent **491**. A side wall of the housing is provided with a second vent, the second vent **492** is in communication with the accommodating cavity, a top of the housing is provided with a third vent **493**, the third vent **493** and the second vent **492** are correspondingly disposed, and the third vent **493** is in communication with the second vent **492**. In this way, through a structural arrangement of the first

draught fan **48** and the second draught fan **49**, an air flow path is formed: the second vent **492**—the fan—the first vent **491**. The third vent **493** is in communication with the second vent **492**, so as to supplement the above-mentioned air flow path and increase an air intake volume in the accommodating cavity. In this way, the air flow rate may be improved, and the flow of the hot air at the second heat exchanger **43** and the radiator **44** is further improved.

[0082] For example, the first heat exchanger **41** is a plate type heat exchanger.

[0083] The embodiments of the present application provide an energy storage box, including a box body, a battery pack, a power conversion system and a thermal management system. The box body includes a battery cabin, the battery pack and the power conversion system are located in the battery cabin. The power conversion system is electrically connected with the battery pack. The battery pack has a first fluid path. The power conversion system has a second fluid path. The thermal management system is partially located outside the battery cabin, and the thermal management system includes a first loop, a second loop and a monitoring unit. The first loop includes a first heat exchanger, a compressor and a second heat exchanger. The first heat exchanger, the compressor and the second heat exchanger are communicated head-to-tail in sequence. The first heat exchanger and the first liquid path are correspondingly disposed. A refrigerant flows in the first loop. The compressor is configured to control the refrigerant to flow in the first loop so as to dissipate heat of a preset liquid in the first liquid path by the refrigerant. The second loop includes a radiator and a solenoid valve, the second liquid path is in communication with the radiator, the first liquid path and the radiator are connected in parallel to the second liquid path, and the solenoid valve is disposed between the second liquid path and the first liquid path. The monitoring unit includes a temperature sensor and a controller, the temperature sensor is electrically connected with the controller, the temperature sensor is configured to monitor a temperature outside the box body, and the controller is electrically connected with the solenoid valve and the compressor. The controller is configured to drive the solenoid valve when the temperature is equal to a preset temperature value, so that a part of the preset liquid in the second liquid path flows through the first liquid path. Through such a structural arrangement, the thermal management efficiency of the energy storage box is improved, and the energy consumption is reduced at the same time.

[0084] It should be noted that, references in the specification to “one embodiment”, “embodiment”, “exemplary embodiment”, “some embodiments” etc., indicate that the described embodiment may include a particular feature, structure, or characteristic, but each embodiment may not necessarily include the particular feature, structure, or characteristic. Furthermore, such phrases are not necessarily referring to the same embodiment. Furthermore, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is within the knowledge of those skilled in the art to implement such feature, structure, or characteristic in connection with other embodiments that are explicitly or implicitly described.

[0085] In general, terms should be understood at least partially from usage in the context. For example, depending at least in part on the context, the term “one or more” as used herein may be used to describe any feature, structure, or characteristic in the singular, or may be used to describe a combination of features, structures, or characteristics in the plural. Similarly, depending at least in part on the context, terms such as “a” or “the” may also be understood as conveying singular usage or plural usage.

[0086] It should be easily understood that, “on”, “above” and “over” in the present application should be interpreted in the broadest way, so that “on” not only means “directly on something”, but also includes “on something” with an intermediate feature or layer therebetween. Further, “above” or “over” not only includes the meaning of “above something” or “over”, but also includes “above something” or “over” without an intermediate feature or layer (namely, directly on something) therebetween.

[0087] Furthermore, for the convenience of description, spatially relative terms, such as “below”,

“under”, “beneath”, “above”, “top” and the like, may be used herein to describe the illustrated relationship of one element or feature with respect to other element(s) or feature(s). Spatially relative terms are intended to encompass different orientations of the device in use or operation other than the orientation shown in the drawings. The device may have other orientations (rotated 90 degrees or in other orientations), and the spatially relative descriptors used herein may also be interpreted accordingly.

[0088] Finally, it should be noted that, the foregoing embodiments are merely intended for illustrating the technical solutions of the present disclosure, but not intended to limit it. Although the present disclosure is described in detail with reference to the foregoing embodiments, it should be understood by those skilled in the art that, the technical solutions described in the foregoing embodiments may still be modified, or some or all technical features thereof may be replaced by equivalents. However, these modifications or replacements do not make the essence of the corresponding technical solutions depart from the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. An energy storage box, comprising a box body, a battery pack, a power conversion system and a thermal management system; wherein the box body comprises a battery cabin, the battery pack and the power conversion system are located in the battery cabin, the power conversion system is electrically connected with the battery pack, the battery pack has a first liquid path, and the power conversion system has a second liquid path; wherein the thermal management system is partially located outside the battery cabin, the thermal management system comprises a first loop, a second loop and a monitoring unit; the first loop comprises a first heat exchanger, a compressor and a second heat exchanger; the first heat exchanger, the compressor and the second heat exchanger are communicated head-to-tail in sequence, the first heat exchanger and the first liquid path are correspondingly disposed; a refrigerant flows in the first loop, the compressor is configured to control the refrigerant to flow in the first loop so as to dissipate heat of a preset liquid in the first liquid path by the refrigerant; wherein the second loop comprises a radiator and a solenoid valve, the second liquid path is in communication with the radiator, the first liquid path and the radiator are connected in parallel to the second liquid path, and the solenoid valve is disposed between the second liquid path and the first liquid path; wherein the monitoring unit comprises a temperature sensor and a controller, the temperature sensor is electrically connected with the controller, the temperature sensor is configured to monitor a temperature outside the box body, the controller is electrically connected with the solenoid valve and the compressor; the controller is configured to drive the solenoid valve when the temperature is equal to a preset temperature value, so that a part of the preset liquid in the second liquid path flows through the first liquid path.
2. The energy storage box according to claim 1, wherein when the temperature is less than the preset temperature value, the controller controls the refrigerant to stop flowing, and drives the solenoid valve, so that a part of the preset liquid in the second liquid path flows through the first liquid path; the preset temperature value is 0 degrees Celsius.
3. The energy storage box according to claim 1, wherein there are at least two radiators, and the at least two radiators are connected in parallel to the second liquid path; there are at least two solenoid valves, one of the at least two solenoid valves and one of the at least two radiators are correspondingly disposed, and one of the at least two solenoid valves is connected in series between the second liquid path and corresponding radiator.
4. The energy storage box according to claim 1, wherein the first loop further comprises an oil separator and a gas-liquid separator, the oil separator is disposed between the compressor and the second heat exchanger; and the compressor, the oil separator and the second heat exchanger are sequentially communicated; the gas-liquid separator is disposed between the first heat exchanger

and the compressor; and the first heat exchanger, the gas-liquid separator and the compressor are sequentially communicated.

5. The energy storage box according to claim 4, wherein there are at least two second heat exchangers, and the at least two second heat exchangers are connected in parallel between the compressor and the first heat exchanger.

6. The energy storage box according to claim 2, wherein the first loop further comprises an oil separator and a gas-liquid separator, the oil separator is disposed between the compressor and the second heat exchanger; and the compressor, the oil separator and the second heat exchanger are sequentially communicated; the gas-liquid separator is disposed between the first heat exchanger and the compressor; and the first heat exchanger, the gas-liquid separator and the compressor are sequentially communicated.

7. The energy storage box according to claim 6, wherein there are at least two second heat exchangers, and the at least two second heat exchangers are connected in parallel between the compressor and the first heat exchanger.

8. The energy storage box according to claim 3, wherein the first loop further comprises an oil separator and a gas-liquid separator, the oil separator is disposed between the compressor and the second heat exchanger; and the compressor, the oil separator and the second heat exchanger are sequentially communicated; the gas-liquid separator is disposed between the first heat exchanger and the compressor; and the first heat exchanger, the gas-liquid separator and the compressor are sequentially communicated.

9. The energy storage box according to claim 8, wherein there are at least two second heat exchangers, and the at least two second heat exchangers are connected in parallel between the compressor and the first heat exchanger.

10. The energy storage box according to claim 1, wherein the box body further comprises a device control cabin, the device control cabin and the battery cabin are disposed at intervals along a length direction of the box body, the compressor is located in the device control cabin, and a top of the device control cabin is in communication with the outside of the box body.

11. The energy storage box according to claim 10, wherein the second heat exchanger and the radiator are disposed at intervals along the length direction of the box body at an outer top of the box body.

12. The energy storage box according to claim 11, wherein the thermal management system further comprises a first draught fan and a second draught fan, the first draught fan and the second heat exchanger are correspondingly disposed, the first draught fan is connected with a top of the box body, and the first draught fan is configured to dissipate heat of the second heat exchanger; the second draught fan and the radiator are correspondingly disposed, the second draught fan is connected with the top of the box body, and the second draught fan is configured to dissipate heat of the radiator.

13. The energy storage box according to claim 12, wherein at least one of the first draught fan and the second draught fan comprises a housing and a fan, the housing has an accommodating cavity, a top of the accommodating cavity has a first vent, and the fan is located in the accommodating cavity so as to discharge air in the accommodating cavity along the first vent; a side wall of the housing is provided with a second vent, the second vent is in communication with the accommodating cavity; a top of the housing is provided with a third vent, the third vent and the second vent are correspondingly disposed, and the third vent is in communication with the second vent.

14. The energy storage box according to claim 2, wherein the box body further comprises a device control cabin, the device control cabin and the battery cabin are disposed at intervals along a length direction of the box body, the compressor is located in the device control cabin, and a top of the device control cabin is in communication with the outside of the box body.

15. The energy storage box according to claim 14, wherein the second heat exchanger and the

radiator are disposed at intervals along the length direction of the box body at an outer top of the box body.

16. The energy storage box according to claim 15, wherein the thermal management system further comprises a first draught fan and a second draught fan, the first draught fan and the second heat exchanger are correspondingly disposed, the first draught fan is connected with a top of the box body, and the first draught fan is configured to dissipate heat of the second heat exchanger; the second draught fan and the radiator are correspondingly disposed, the second draught fan is connected with the top of the box body, and the second draught fan is configured to dissipate heat of the radiator.

17. The energy storage box according to claim 16, wherein at least one of the first draught fan and the second draught fan comprises a housing and a fan, the housing has an accommodating cavity, a top of the accommodating cavity has a first vent, and the fan is located in the accommodating cavity so as to discharge air in the accommodating cavity along the first vent; a side wall of the housing is provided with a second vent, the second vent is in communication with the accommodating cavity; a top of the housing is provided with a third vent, the third vent and the second vent are correspondingly disposed, and the third vent is in communication with the second vent.

18. The energy storage box according to claim 3, wherein the box body further comprises a device control cabin, the device control cabin and the battery cabin are disposed at intervals along a length direction of the box body, the compressor is located in the device control cabin, and a top of the device control cabin is in communication with the outside of the box body.

19. The energy storage box according to claim 18, wherein the second heat exchanger and the radiator are disposed at intervals along the length direction of the box body at an outer top of the box body.

20. The energy storage box according to claim 1, wherein the first heat exchanger is a plate type heat exchanger.
